SevaOne - Service Plus

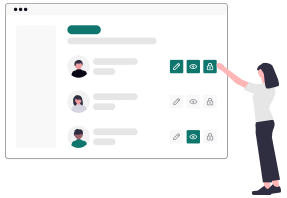
Functional Architecture

The lifecycles we will build, and the understanding we still need

Proposal for discussion

What We Will Build

Twenty-five features, in five categories. The pages that follow show how each works.



People and offices

M1
Accounts and sign-in
registration, one-time codes, officer onboarding

M2
Offices and areas
departments, offices, districts, dated postings

M3
Roles and access
who may see and do what

M13
Operators and centres
assisted applications at service centres



Service setup

M4
Service catalogue
the list of services and how each is launched

M5
Form builder
drag-and-drop forms in many languages

M6
Rules and fees
papers needed and fees, set by tables

M17
Publishing and versions
test, approve, go live, change safely

M22
Derived services
one service offering a follow-on service

M25
Service search
finding the right service in your language



Application work

M7
Workflow
steps, choices, parallel reviews, deadlines

M8
Applications
drafts, submit once, track, correct

M9
Officer work
work lists, claim, decide

M21
Bulk processing
act on many cases at once

M24
Time limits
clocks, reminders, escalation

M15
Complaints and appeals
up to three levels



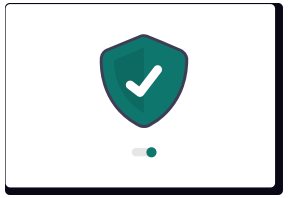
Documents and money

M10
Certificates
made, signed, issued, checked by QR

M11
Payments
fees, receipts, matching with the bank

M12
Messages
SMS, email and WhatsApp

M23
Benefit payments
paying a benefit after approval



Platform and support

M14
Outside connections
identity, signing, payments, departments

M16
Reports
dashboards and exports by role

M18
History and operations
every action recorded, monitoring, recovery

M19
Platform settings
name, languages, gateways, calendar

M20
Moving Karnataka
bringing existing services and applications across

Illustrations: unDraw.

How to Read This

This is our proposal. Every page shows one process from start to finish, what we will build, and where we still need understanding.

Understanding needed

A step or topic where we still need to learn the department's rules or choices before we can finish the design.



The stacked-rings picture marks a place where records are kept.

M4

Small grey tags are module numbers: the parts of the platform that do this step. The key is below.

Module numbers

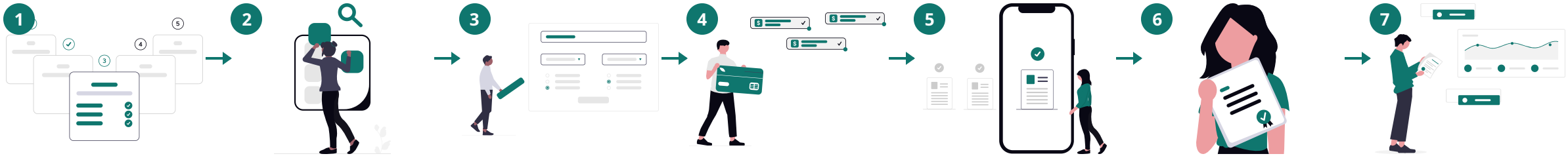
M1 Accounts and sign-in
M2 Offices and areas
M3 Roles and access
M4 Service catalogue
M5 Form builder
M6 Rules and fees
M7 Workflow
M8 Applications
M9 Officer work
M10 Certificates
M11 Payments
M12 Messages
M13 Operators and centres
M14 Outside connections
M15 Complaints and appeals
M16 Reports
M17 Publishing and versions
M18 History and operations
M19 Platform settings
M20 Moving Karnataka
M21 Bulk processing
M22 Derived services
M23 Benefit payments
M24 Time limits
M25 Service search

Illustrations: unDraw.

The Whole Lifecycle in One Picture

Modules 1 · 2 · 3 · 4 · 5 · 6 · 7 · 8 · 9 · 10 · 11 · 12 · 15 · 16 · 17 · 18

From setting up a service to the citizen getting the result, and what stays underneath.



1 Set up the service

The service owner designs the form, steps, papers and messages.

M4 M5 M6 M7 M17

Service library

2 Citizen finds it

The citizen searches or browses the list of services.

M4

3 Citizen applies

Fills the form, adds papers and submits.

M1 M5 M8

Applications

4 Pays the fee

Pays online if the service has a fee.

M11

Payment records

5 Officer decides

The case goes to the right office. The officer reviews and decides.

M2 M3 M7 M9

↩ correction asked: back to the citizen

6 Certificate and message

A certificate is issued and the citizen is told.

M10 M12

Issued copies

7 Afterwards

Appeals, reports and a permanent history of every action.

M15 M16 M18

History

Always underneath: platform settings · who may do what · history of every action · links to other systems

What we will build

- One platform for the whole lifecycle: setting up a service, applying, deciding, paying, certificates and appeals.
- The same lifecycle works for all three kinds of service (see page 13).
- Every step leaves a permanent record that is never edited.

Understanding needed

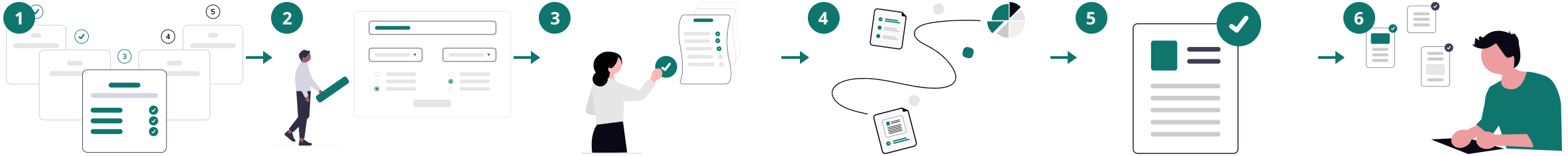
- Which services Karnataka will start with.
- The number of services, applications and users to plan for.
- Hosting, support and recovery arrangements.

Illustrations: unDraw.

Setting Up a Service

Modules 4 · 5 · 6 · 7 · 10 · 12 · 17

How a service owner turns a government service into something citizens can use.



1 Define the service

Name, who can apply, papers, fee, time, and which department owns it.

M4

2 Design the form

Drag and drop fields, pages and languages. Add checks.

M5

3 Set rules and fees

Which papers are needed for whom, and what it costs.

Understanding needed M6

4 Draw the steps

Who handles it, in what order, with what choices.

M7

5 Certificate and messages

Design the certificate and the messages people get.

M10 M12

6 Test, approve, go live

Try it, have a second person approve, then publish.

M17

Service library

↩ a later change makes a new version

What we will build

- Form, rules, steps, certificate and messages are saved together as one version.
- A second person must approve before a service goes live.
- Applications already running keep the version they started with.
- Ordinary changes need no developer. Special scripts are reviewed first.

Understanding needed

- Who approves a launch, and in what order.
- What happens to running applications when a new version is published.
- How much the department wants to change without developer help.
- Which services will be set up first.

Illustrations: unDraw.

People, Offices and Access

Modules 1 · 2 · 3 · 13

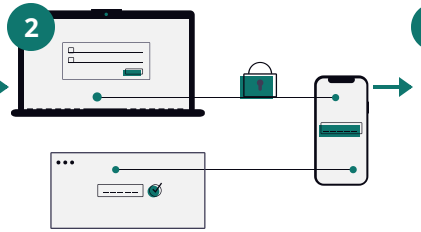
Who is in the system, where they sit in the government, and what they are allowed to do.



1 Person registers

Citizens create an account. Officers are added by an administrator.

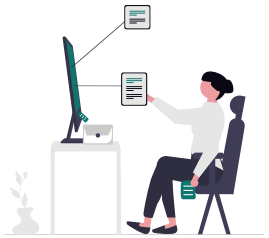
M1



2 Identity is confirmed

A one-time code to the phone or email.

Understanding needed M1



3 Offices and areas

Departments, offices, districts and the areas each one looks after.

M2

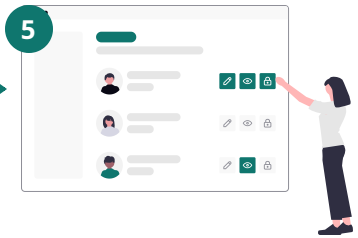
Offices and posts



4 Officer gets a post

The officer is posted to an office from a date.

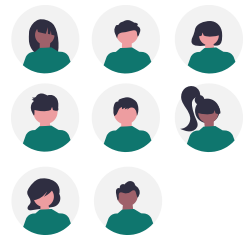
M2



5 Role sets what they can do

Role, office, area and the case itself decide access.

M3



6 Centre operators

Operators at service centres apply for citizens.

Understanding needed M13

What we will build

- One account per person. Citizen work and officer work are separate pages.
- A posting has dates: a transfer never rewrites what the officer did before.
- Access = role + office + area + the case assigned to them. A title alone is not enough.
- Centre operators are recorded alongside the citizen they helped.

Understanding needed

- How officer accounts are approved, suspended and closed.
- Whether one login can be used for both citizen and officer work.
- Which official lists define offices and areas, and how overlaps or gaps are handled.
- How service centres are accredited and what they may offer.

Illustrations: unDraw.

The Citizen Lifecycle

Modules 1 · 4 · 5 · 6 · 8 · 10 · 11 · 12 · 15

What a citizen goes through, from the first visit to getting an answer.



1

Arrives

Opens the website or the mobile app.

M8



2

Account and sign in

Registers once, then signs in.

M1



3

Finds a service

Searches or browses and sees who can apply.

M4



4

Applies and pays

Form, papers, fee, then a receipt number.

M5 M6 M8 M11



5

Waits and tracks

Sees status. Gets a message if something is needed.

M8 M12



6

Gets the result

Downloads the certificate or reads the rejection.

M10



7

If unhappy

Complains or appeals.

M15

Applications

↩ officer asks for a correction

What we will build

- One personal page for everything: applications, payments, certificates and complaints.
- Drafts are saved. A double tap never makes two applications.
- Status is shown in plain words for every kind of service, including ones run by another department.
- Messages by SMS, email or WhatsApp at each important step.

Understanding needed

- How a citizen proves they may apply for someone else (guardian, organisation).
- When the reference number is final, and when the deadline clock starts.
- Languages needed on every screen and message.
- How citizens recover lost access.

Illustrations: unDraw.

The Life of One Application

Modules 7 · 8 · 9 · 10 · 18

The stages an application passes through. Payment and certificate have their own stages, kept apart.



1 Draft

Being filled. Only the citizen sees it.

M8

2 Submitted

Saved. Has a reference number.

M8

Applications

3 With the officer

Being checked and decided.

M9

4 Waiting for citizen

A correction was asked. The same case waits.

M7

5 Approved or rejected

The decision is recorded with a reason.

M7

Case history

6 Certificate issued

Made, signed and sent.

M10

7 Closed

Nothing more to do. History stays.

M18

↩ citizen fixes it, back to the same officer

Payment: not due → due → paid → matched with the bank (or failed → refund) | Certificate: not started → being made → signed → issued

What we will build

- Stages: Draft, Submitted, Under scrutiny, Waiting for applicant, Under verification, Pending decision, Approved, Rejected, Closed.
- A pending payment never means approved. A failed signing never undoes a decision.
- The case keeps the service version it started with.

Understanding needed

- Which stages stop the deadline clock.
- Who may withdraw an application, and when.
- What each stage should be called in the department's own words.

Illustrations: unDraw.

Applying for a Service

Modules 1 · 4 · 5 · 6 · 8 · 12

The citizen's steps in detail, from finding the service to the receipt.

1

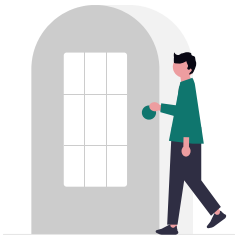


Find the service

Search, browse, or follow a link.

M4

2

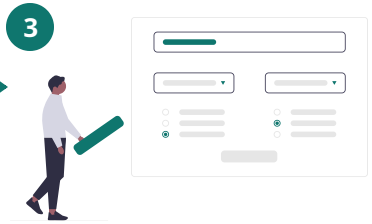


Sign in

Phone or email with a one-time code.

M1

3

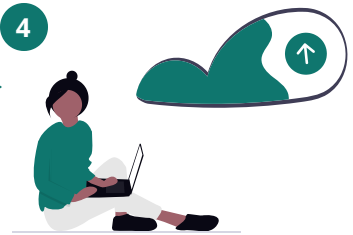


Fill the form

Only the questions that apply. Help text and checks as you go.

M5 M6

4



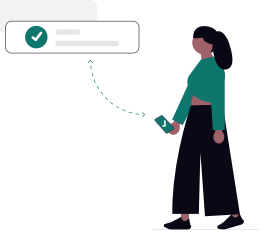
Add papers

Upload what is needed.

Understanding needed M5

Papers

5



Check and submit

Review, then submit once.

M8

Applications

6



Receipt and message

A reference number and an SMS or email.

M8 M12

What we will build

- The form asks only for papers the rules say are needed.
- Each submit carries a one-time key, so a double tap makes one application.
- The receipt is given only after the save has succeeded.
- Checks run on the form and again on the server.

Understanding needed

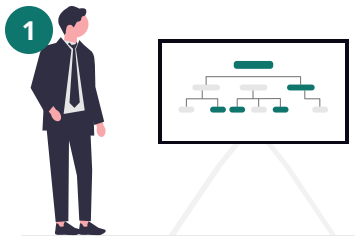
- Allowed paper types, sizes and any scanning for viruses.
- Whether a service takes payment before or after submitting.
- Rules for guardians, organisations and representatives.

Illustrations: unDraw.

How an Officer Handles a Case

Modules 2 · 3 · 7 · 9 · 24

From the case landing in a work list to a decision, including corrections and deadlines.

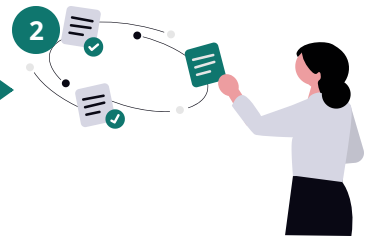


1

Case reaches the right office

By area, and then by the role at that step.

Understanding needed M2 M3



2

Officer opens the work list

Sees only cases they may handle.

M9

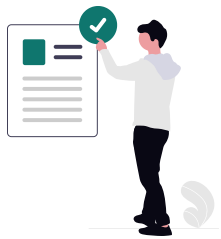


3

Reviews form and papers

Reads earlier remarks and history.

M9

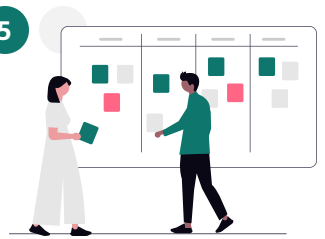


4

Chooses an action

Forward, ask for correction, approve or reject.

M7

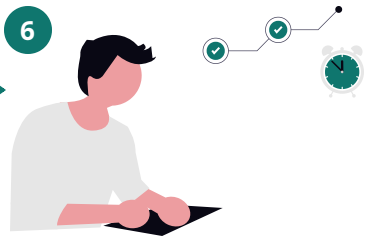


5

More reviews if needed

One after another, or several at the same time.

M7



6

Deadlines watched

Late cases are shown first and escalate.

Understanding needed M24

Case history

↩ correction: back to the citizen

What we will build

- The case goes to office + role, using the area and the current step.
- Every action has a fixed code. The button text never decides what happens.
- Claiming a case stops two officers working on it.
- Late cases are listed first and escalate automatically.

Understanding needed

- How one officer is chosen when several share a role, or when one is absent.
- How long a claim lasts and who may release it.
- Which decisions need a written reason.
- How officers are told about new cases.

Illustrations: unDraw.

Payments

Modules 6 · 11 · 14

Money is handled as four separate things so that mistakes cannot approve a case or charge twice.



1 **Fee worked out**

From the rules for this service and this person.

Understanding needed M6 M11

2 **Citizen pays online**

Through a payment gateway.

M11

3 **Gateway replies**

A signed message says paid or failed.

M11 M14

4 **Matched with the bank**

A daily bank file confirms it.

M11

5 **Receipt shown**

Only after the match.

M11

6 **Problems**

Refunds, duplicates and unmatched payments.

Understanding needed M11

Payment records

↩ no reply: ask the gateway again

What we will build

- Four separate things: amount due, attempt, receipt and money received.
- A browser return or a screenshot is never proof of payment.
- A repeated callback is ignored by transaction number. Payment never approves a case.
- If saving the application fails after payment, the money stays tied to the application.

Understanding needed

- Which payment gateways and accounts will be used.
- Settlement times, refunds and reversals.
- Who owns matching and finance problems.
- Whether payment comes before or after submitting.

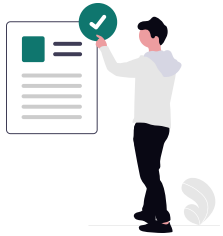
Illustrations: unDraw.

Certificates and Messages

Modules 7 · 10 · 12 · 14

What the citizen receives once a decision is made.

1



Decision made

Approved or rejected, with a reason.

M7

2



Certificate made

Template filled with the case data.

M10

3



Signed

A digital signature from the authorised person.

Understanding needed M10

M14

4



Issued with number and QR

Stored exactly as issued.

M10

Issued copies

5



Citizen told

SMS, email or WhatsApp.

Understanding needed M12

6



Anyone can check it

Scan the QR, see the agreed details.

M10

What we will build

- The issued copy is kept exactly as issued. A correction makes a new record.
- A failed signature leaves the certificate "pending" and retries. A failed message never undoes a decision.
- The check page shows only agreed details, with no private data.

Understanding needed

- Who signs each certificate, and with what authority.
- Validity, cancellation and correction rules for each certificate.
- Which message channels, senders and consent rules apply.

Illustrations: unDraw.

Three Kinds of Service

Modules 4 · 8 · 14

A service entry alone does not tell the app how to run it. Each service carries its type.

A. Done inside the platform



Form
Citizen fills it.



Officers
Work it through the steps.

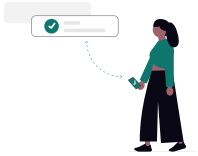


Certificate
Issued here.

B. Collection only



Form
Citizen fills it.



Number
An application number is given.



Done
No approval chain.

C. Handed to a department



Token
A short-lived signed token is made.

Understanding needed



Department site
Form and payment happen there.



Result back
A signed message returns the result.

Understanding needed

What we will build

- The catalogue records each service's type, so every screen knows how to launch it.
- One status page works for all three kinds.
- Handover uses a short-lived signed token and a signed callback.
- If the citizen stops halfway, the status shows "with department" and can be resumed.

Understanding needed

- Who makes the token, its format, how long it lasts, and how the other department checks it.
- How the callback is secured, and what happens if it is repeated or never arrives.
- What a citizen sees if they stop halfway, and how logout works.

Illustrations: unDraw.

Complaints and Appeals

Modules 2 · 12 · 15

What a citizen can do when they are not happy with the service or the decision.

1



Citizen raises it

A complaint, or an appeal against a decision.

M15

2

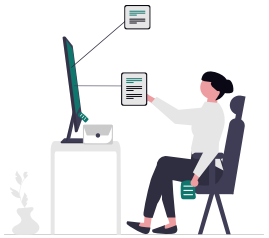


Linked to the case

The original application is attached.

M15

3



Goes to the right authority

Chosen by the rules for this service.

Understanding needed M2

M15

4



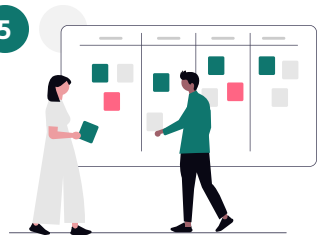
Authority decides

Gives a reason.

M15

Case history

5



Escalates if needed

Up to three levels.

M15

6



Citizen told

At every step.

M12

What we will build

- A complaint, an appeal and a technical support ticket are three different things.
- An appeal never changes the original case record.
- Deadlines escalate automatically up to three levels.

Understanding needed

- Which law and procedure apply to appeals in Karnataka.
- Who can appeal, within how many days, and whether hearings are needed.
- Who may see which papers.

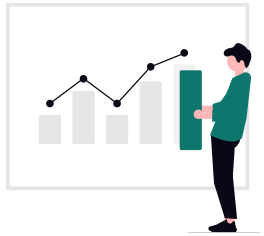
Illustrations: unDraw.

Reports, History and Keeping It Running

Modules 16 · 18

How the government sees what is happening, and how problems are caught.

1



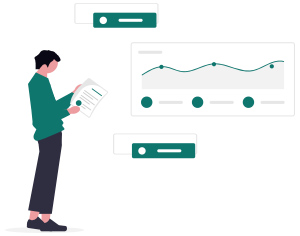
Every action recorded

Written once, never edited.

M18

History

2



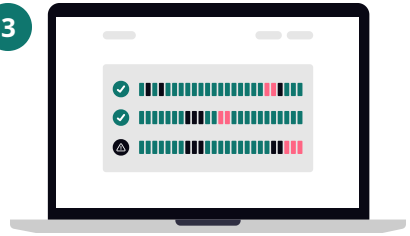
Reports from a copy

So reports never slow down live work.

M16

Reporting copy

3



Dashboards by role

Each person sees only what they should.

Understanding needed M16

4

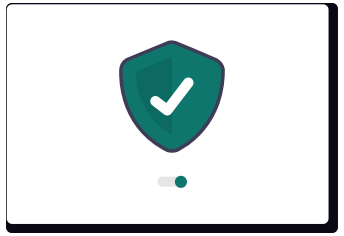


Alarms

Failed jobs and broken connections reach support.

M18

5



Backups and recovery

Copies kept apart, restores practised.

Understanding needed M18

What we will build

- One history of every decision, assignment, setting change and sign-in.
- Monitoring, alarms and rehearsed recovery.
- Reports use agreed definitions of "pending", "disposed" and "overdue".

Understanding needed

- The agreed meaning of pending, disposed, overdue and completed.
- Who sees named records and who sees only totals.
- How long history is kept, and who handles incidents and citizen support.

Illustrations: unDraw.

Connections to Other Systems

Modules 14

Everything outside the platform goes through one connection point.



Identity

Aadhaar, DigiLocker

Understanding needed



Signing

eSign

Understanding needed



Payments

Gateways and treasury

Understanding needed



Connection point retries · logs · named owner



Benefit money

PFMS and DBT

Understanding needed



Messages

SMS, email, WhatsApp, Sandes

Understanding needed



Departments

APIs, callbacks, handover

Understanding needed

What we will build

- Every outside link has a named owner, agreed fields, a timeout, retries and a log.
- Departments reach us through the API manager with their own keys and limits.
- No service talks to an outside system directly.

Understanding needed

- Which connections Karnataka needs, and who owns each agreement.
- Which fields may be exchanged, and on what basis.
- Test access and approval for each connection.

Illustrations: unDraw.

Platform Settings

Modules 2 · 3 · 5 · 11 · 12 · 19 · 24

The settings that make the platform Karnataka's own.

1

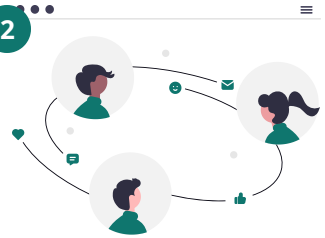


Name and branding

Karnataka's name, logo and web address.

Understanding needed M19

2

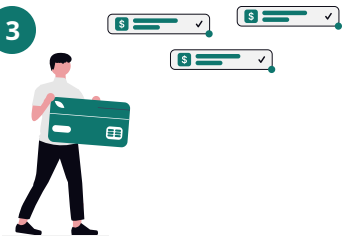


Languages

Kannada, English and Hindi on every screen and message.

Understanding needed M5 M19

3

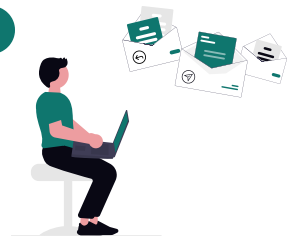


Payment gateways and accounts

Which gateways and bank accounts are switched on.

Understanding needed M11 M19

4

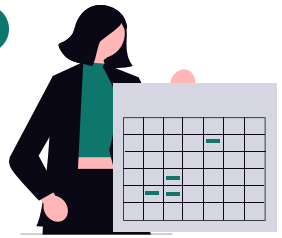


Message senders and templates

The SMS, email and WhatsApp senders and wording.

Understanding needed M12 M19

5

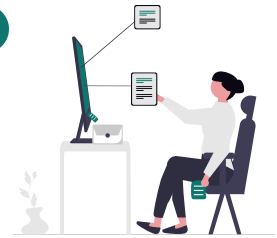


Working days and holidays

The calendar used for deadlines.

Understanding needed M24 M19

6



Offices and roles

Departments, offices and who may do what.

M2 M3

What we will build

- One settings area for the whole platform, with every change recorded.
- Sensitive settings, such as payments and signing, need a second person to approve.
- Platform support can reach records only with approval, and every such access is logged.

Understanding needed

- The official name, branding and web address.
- Which languages are needed, and who approves the wording.
- Which gateways and message senders will be used.
- The holiday calendar and working-day rules.

Illustrations: unDraw.

Moving Karnataka to the New Platform

Modules 20

Existing services and applications must come across, with their numbers and certificates still working.



1 Get the export

From the current system.

Understanding needed

M20

2 Copy it untouched

Into a staging area.

M20

Staging copy

3 Map to the new model

Services, forms, steps, cases.

Understanding needed

M20

4 Report what does not fit

Never silently dropped.

M20

5 Load

Services, applications, papers, certificates.

M20

New platform data

6 Compare

Counts and samples must match.

M20

7 Switch over

With a way back.

Understanding needed

M20

↩ repeat the rehearsal until the numbers match

What we will build

- A staging area, and a loader that can be run again safely.
- Old application numbers and certificate QR checks keep working.
- Open cases restart at the right step, not from the beginning.
- Rehearse until the numbers match.

Understanding needed

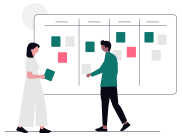
- Who may release the export, in what form, and when.
- Whether papers, signatures, payment records and history are all included.
- How open cases map to the new steps.
- How a rollback keeps decisions and payments made after the switch.

Illustrations: unDraw.

Five More Features

Modules 21 · 22 · 23 · 24 · 25

These sit alongside the first twenty modules.



Bulk processing

Officer picks many cases → one job each → each acted on → report

- **We will build:** Same checks as a single case. One failure never stops the rest.
- **Understanding needed:** Which actions can be done in bulk, and the limits.



Derived services

Case approved → rule offers a follow-on service → form prefilled

- **We will build:** The two applications stay linked.
- **Understanding needed:** Which services lead to which follow-on services.



Benefit payments

Approved → account checked → release order → status matched

- **We will build:** Kept apart from fee money. Paid once per approval.
- **Understanding needed:** Which programmes are covered, and who owns sanction and release.



Time limits

Limit set → clock starts → late → reminder and escalation

- **We will build:** A working-day calendar. The citizen sees the due date.
- **Understanding needed:** Clock rules, pauses, and who receives escalations.



Service search

Citizen types a word → finds services → opens one

- **We will build:** Search in the citizen's language. Retired services never shown as open.
- **Understanding needed:** Languages to support and how services are described.

Illustrations: unDraw.

Understanding Needed, Module by Module

Part 1 of 3. Topics we still need to understand with the department.

M1 Accounts and sign-in

- Approval, activation and suspension of officer accounts.
- One login for citizen and officer work.
- Recovery when a phone or email is lost.

M2 Offices and areas

- The official lists that define offices and areas.
- Overlapping or missing areas.
- Effect of transfers and boundary changes on cases in progress.
- Who drafts, checks and activates changes.

M3 Roles and access

- The full list of permissions and separation of duties.
- Delegation, temporary charge and access across offices.
- Access when an officer is absent.

M4 Service catalogue

- The full Karnataka list of services and their types.
- Legal ownership of each service and who may apply.
- The official wording of each service.

M5 Form builder

- Field types, upload limits and the accessibility standard.
- Approval of Kannada, English and Hindi wording.
- How special scripts are controlled.

M6 Rules and fees

- Fee, exemption and rounding rules for each service.
- Which date fixes the fee, and which changes need a new assessment.
- Who owns the rule tables.

M7 Workflow

- The action codes and their exact meaning.
- Calendars, pause rules and escalation authority.
- Parallel reviews that disagree or are cancelled.
- Retry rules for failed outside calls.

M8 Applications

- Guardians, organisations and representatives.
- When the reference is final and when the clock starts.
- Drafts, withdrawal and corrections.

M9 Officer work

- How long a claim lasts, and who may release it.
- Reasons and review rights for decisions.
- How officers are told about new cases.

M10 Certificates

- Who signs, with what authority, and the seal policy.
- Validity, cancellation and correction rules.
- What a verification page may show.

Illustrations: unDraw.

Understanding Needed, Module by Module

Part 2 of 3. Topics we still need to understand with the department.

M11 Payments

- Collection heads, settlement owners and the refund process.
- Timeouts, double debits and part payments.
- Payment that works when the application is not saved.
- Whether payment comes before or after submitting.

M12 Messages

- Approved channels, senders, templates and consent rules.
- Fallback language, and failed or changed contacts.
- Which channels are switched on.

M13 Operators and centres

- Accreditation of centres and what they may offer.
- How a citizen authorises help, and when consent is asked again.
- Wallets, commissions and settlement.

M14 Outside connections

- Which connections Karnataka needs, and who owns each.
- Which fields may be exchanged and on what basis.
- Token maker, format, lifetime and callback security for handed-over services.

M15 Complaints and appeals

- The law and procedure that apply.
- Who can appeal, how soon, and whether hearings are needed.
- How a complaint differs from a correction and a formal appeal.

M16 Reports

- The agreed meaning of pending, disposed, overdue and completed.
- Who sees named records and who sees only totals.
- How fresh the numbers must be.

M17 Publishing and versions

- Who approves policy, test results and launch.
- Drafts and running cases when a version changes.
- Rules for major and minor versions.

M18 History and operations

- How long history is kept, and who may read or delete it.
- Who handles incidents, citizen support and escalation.
- Availability, recovery and backup targets.

M19 Platform settings

- The official name, branding and web address.
- Languages needed, and who approves the wording.
- Gateways and message senders to be used.
- The holiday calendar and working-day rules.

M20 Moving Karnataka

- Who may release the export, and in what form.
- Whether papers, signatures, payments and history are all included.
- How open cases are mapped, and how rollback works.

Illustrations: unDraw.

Understanding Needed, Module by Module

Part 3 of 3. Topics we still need to understand with the department.

M21 Bulk processing

- Which actions may be done in bulk, and the limits.

M22 Derived services

- Which services lead to which follow-on services.

M23 Benefit payments

- Programmes covered, and who owns sanction and release.

M24 Time limits

- Clock rules, pauses and who receives escalations.

M25 Service search

- Languages to support and how services are described.

Illustrations: unDraw.

Understanding Needed First

These topics change the design the most.

1. **Existing data**
   Access to Karnataka's current service settings and applications, and their format.

2. **First services**
   The real services to map first: one simple, one with several reviews, one with a fee and a certificate.

3. **Outside connections**
   Which identity, signing, payment and benefit connections are needed, and who owns each.

4. **Action codes**
   The exact meaning of each action an officer can take.

5. **Payment failures**
   What happens when money is taken but the application is not saved.

6. **Signing**
   Who signs certificates, and how the signature is checked.

7. **Policy owners**
   Who approves service rules, fees, launches and changes.

8. **Appeals**
   The law that governs appeals and their time limits.

9. **Hosting and support**
   Where the platform will run, and who supports it day to day.

10. **Earlier conversations**
    Points raised in recorded discussions that have not been written down yet.

Illustrations: unDraw.